Data 3 HW

Consolidation

1 i) Positive Correlation
ii) Two different data sets

2 Correlation coefficient becomes weaker since linear relationship becomes less visible

3 i) A) number of minutes per mile
B) Initial time for a journey

ii) A) $y = 0.5 \times 0.7 + 9$
$= 9.35$
B) $y = 7 + 9 = 16$
C) $y = 140 + 9 = 149$ miles

iii) A: no close data ∴ extrapolation
B: close to data → strong correlation: accurate
C: Far from the dataset so not accurate

4 a) The higher the life expectancy in 1970, the higher the life expectancy in 2000

b) $\frac{64-53}{16} = 0.69$

c) coefficient less than Asian one because points are further out from line of best fit

d) both have generally positive correlations

Exam Question

1

a) 0.913

-0.913 and -0.124 are both negative; not the coefficient

0.124 is ~~too~~ close to 0,

higher positive correlation than this.

b) correlation ≠ causation. The relationship is more likely to be due to population sizes.

c) population sizes are not fixed; relationship is due to increasing population sizes.